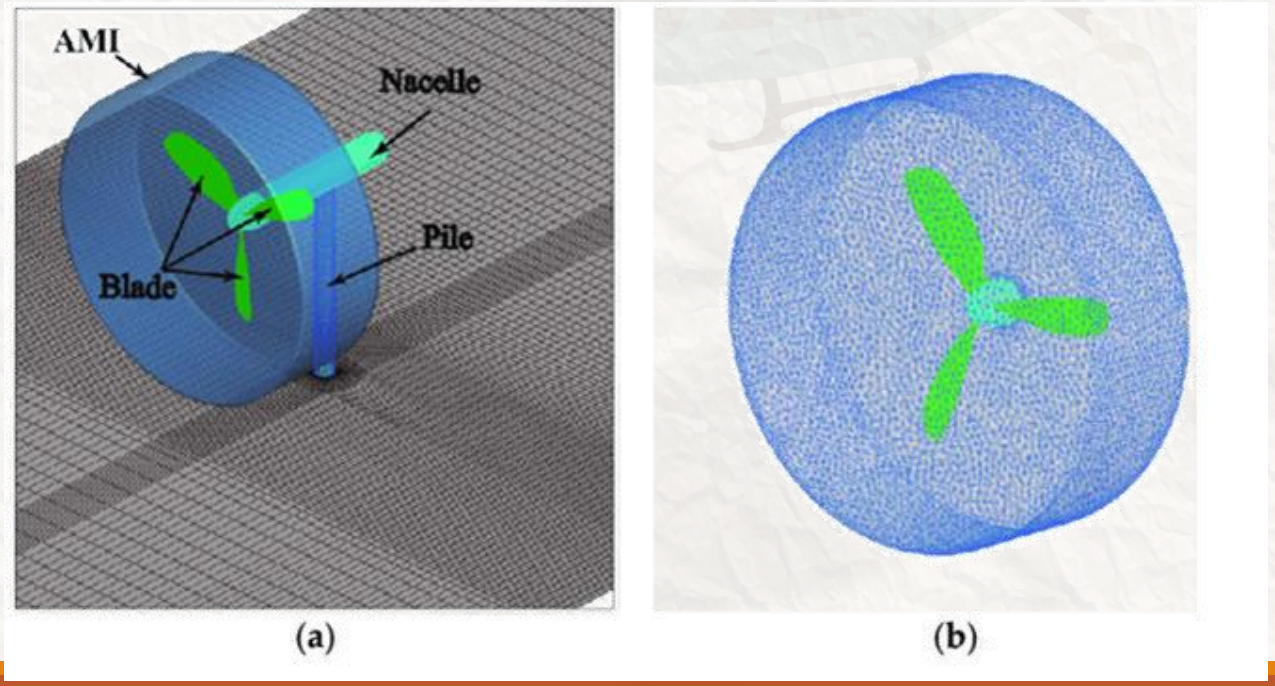
Rotary Wing Aerodynamics (AE 667)



Types of Rotor Simulation Tools

Finite Volume Method type

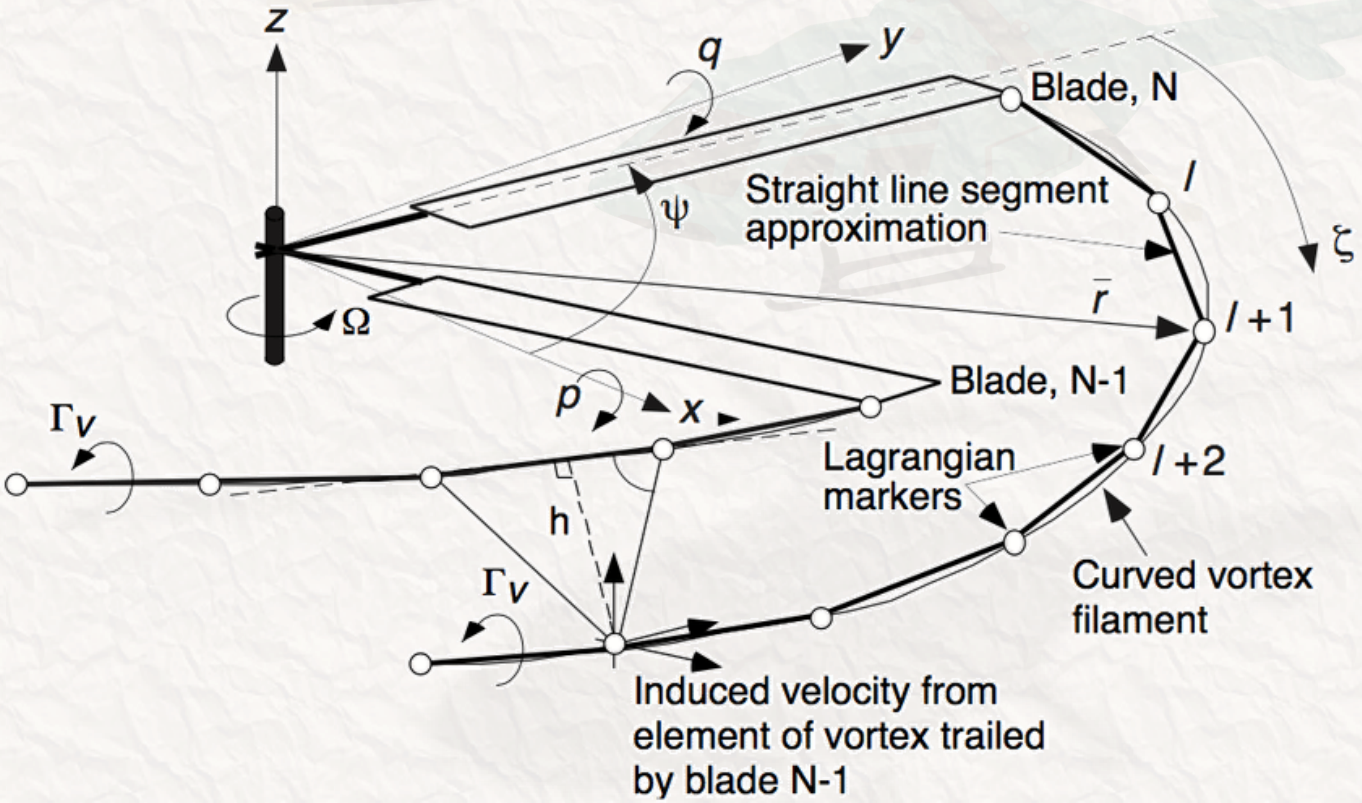
- Mesh based Navier Stokes CFD.
- Rotating and Static fluid domains



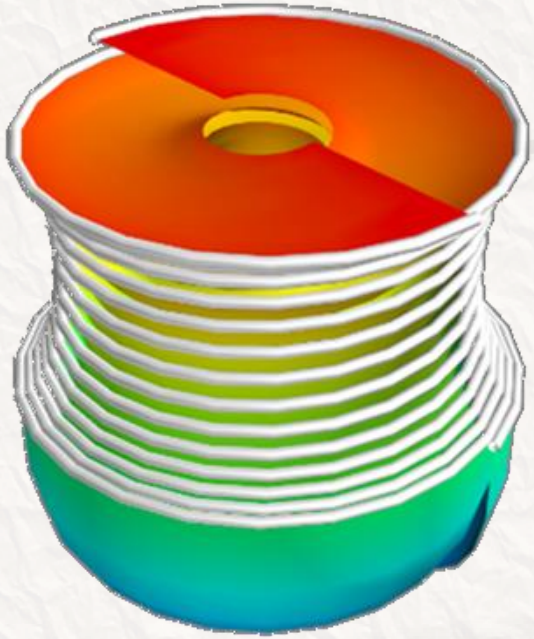
https://www.researchgate.net/publication/331851708_Comparison_of_Actuator_Line_Method_and_Full_Rotor_Geometry_Simulations_of_the_Wake_Field_of_a_Tidal_Stream_Turbine

Vortex Method Type

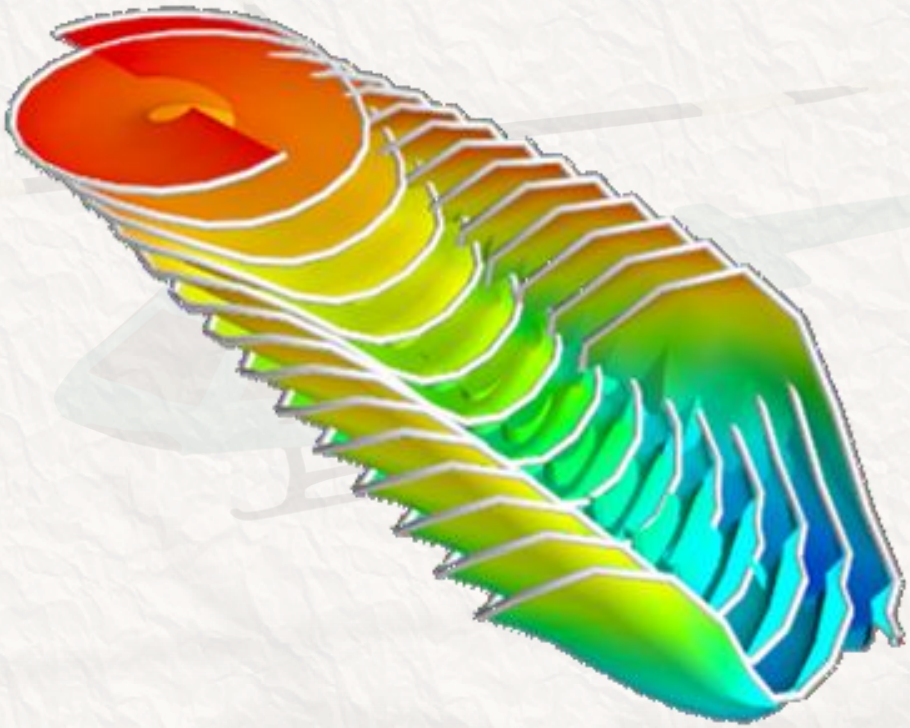
- Lifting Line/Surface with Free Wake Method



Vortex Method In-house Code Development (Sharookh)

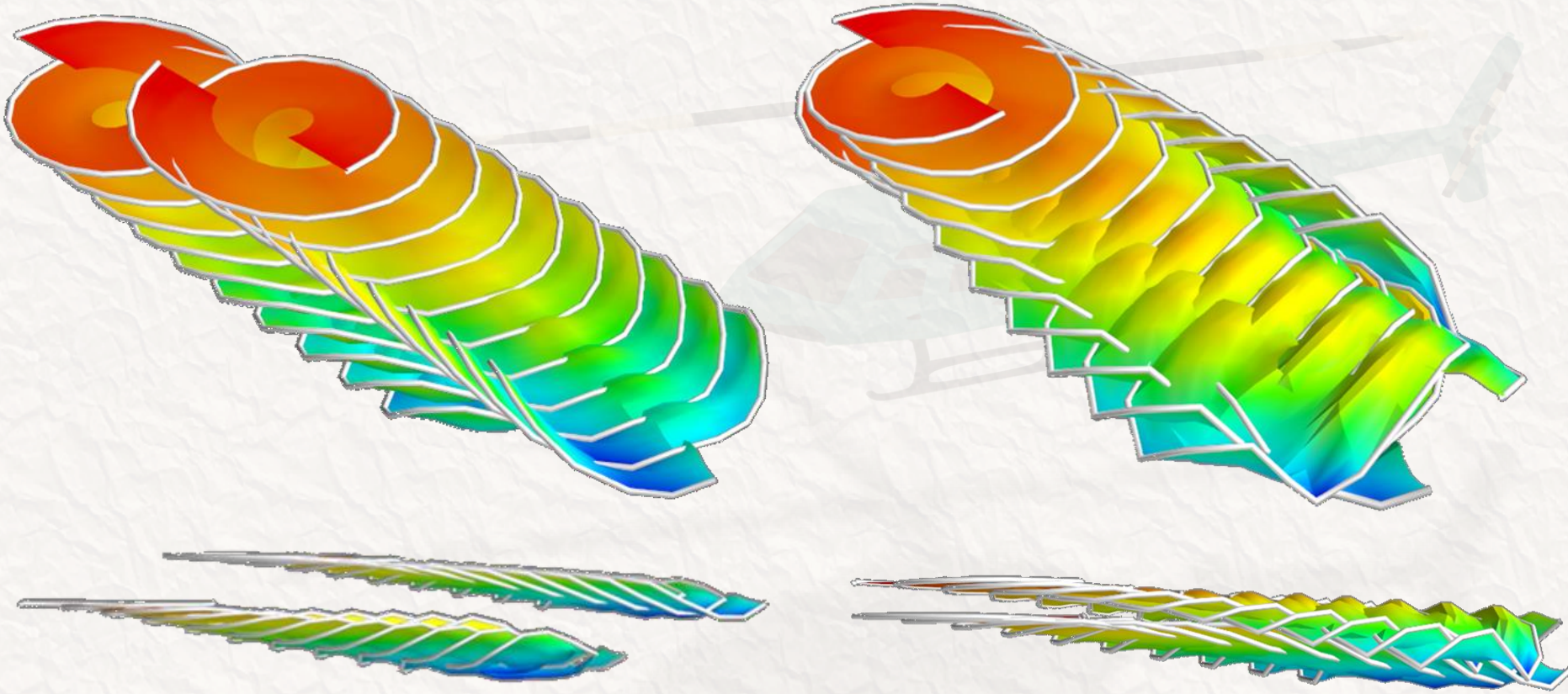


V= 0 m/s

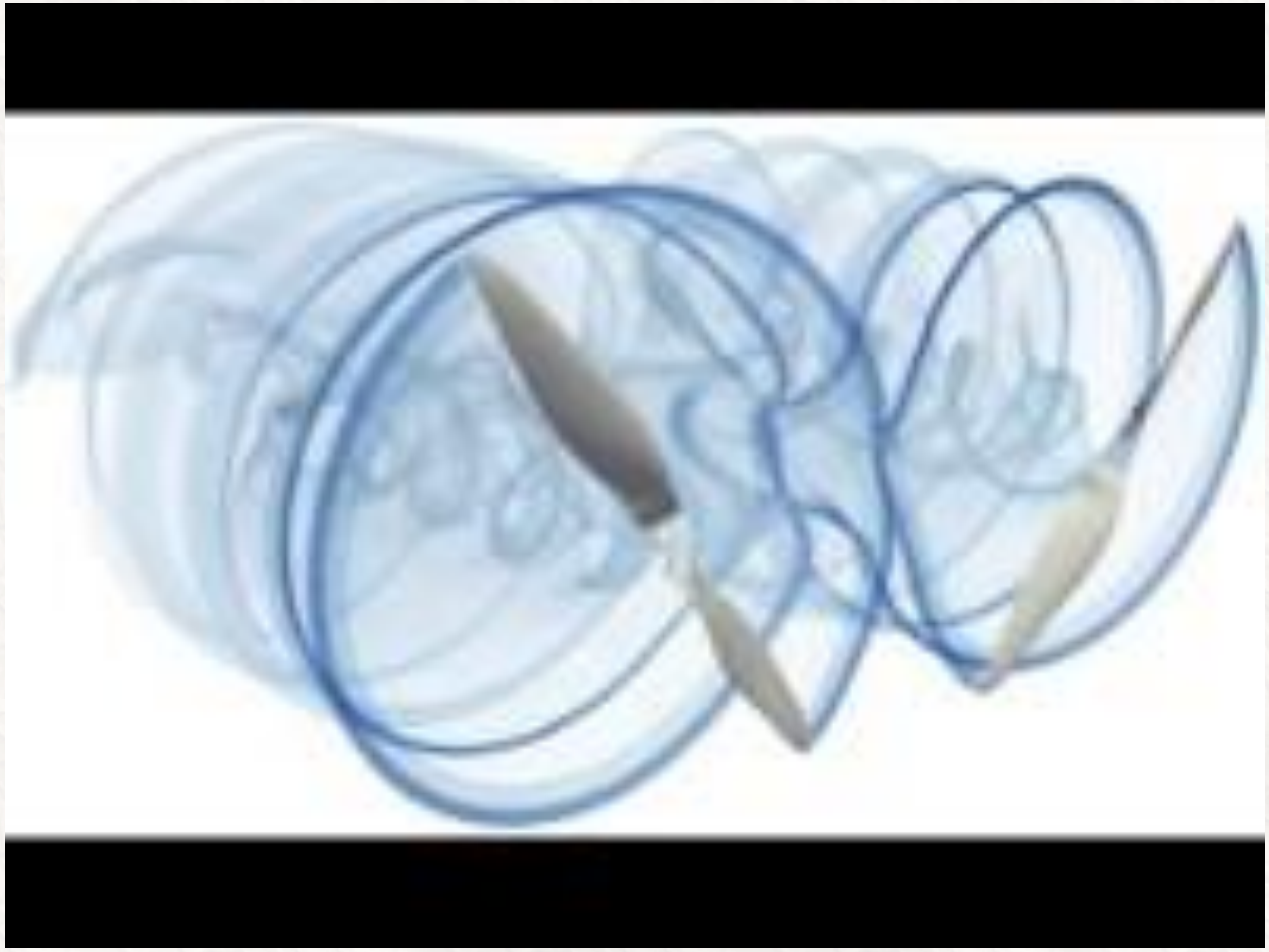


V= 20 m/s

Vortex Method In-house Code Development (Sharookh)

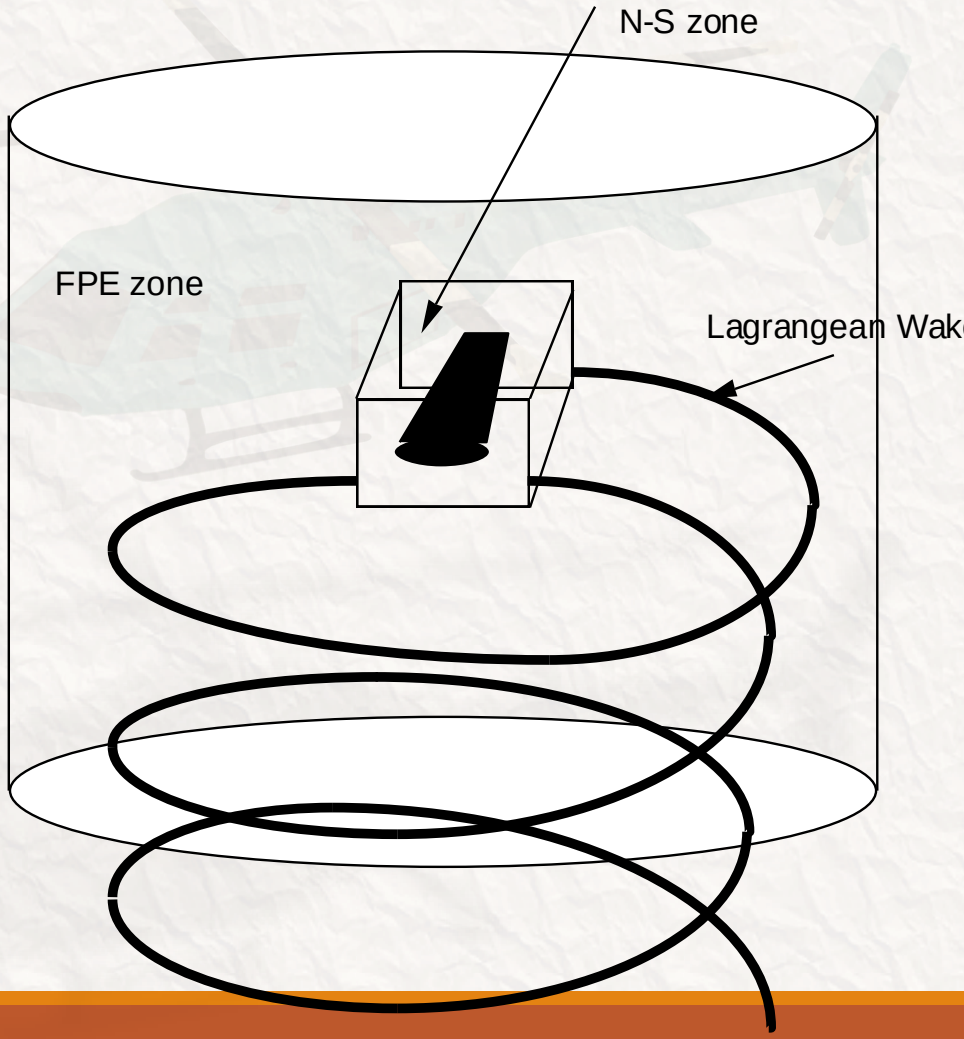


Viscous Vortex Particle Method



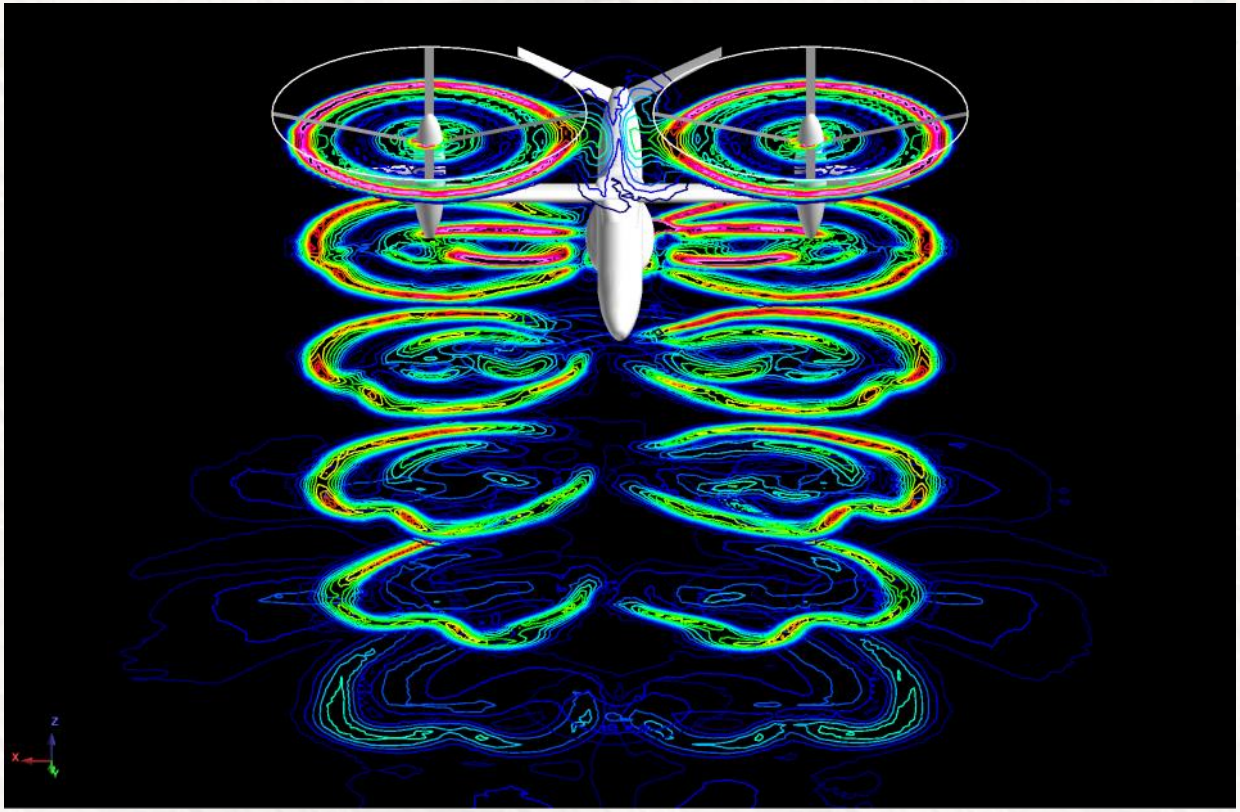
Hybrid Type - 1

- N-S Zone
- Potential Zone
- Lagrangian Wake



Hybrid Type - 2

- Rotor blades modeled as momentum sources
- N-S in wake



Reflections…



https://forms.gle/7Fi8rZYJprUxHeVc6